# Marlenne Esparza

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## EDUCATION

Texas A&M University – College Station, TX *May 2026*
Bachelor of Science in Multidisciplinary Engineering Technology – Mechatronics ***GPA***: 3.14/4.00
***Minors:*** Embedded Systems Integration, Mathematics

## SKILLS

***Robotics & Automation:*** ROS (RViz, MoveIt 2), UR Cobot Programming (Polyscope)
***Programming:*** Python, C/C++, Assembly, Kotlin, React/Next.js, Linux
***Instrumentation & Testing:*** Oscilloscope, Multimeter, Signal Generator, AC/DC Power Supply
***Hardware & Electronics:*** PCB Design, KiCad, Multisim, Quartus, LiDAR, Node-RED, SPI, I2C, UART
***Modeling & Simulation***: SolidWorks, Autodesk Fusion 360, Onshape, MATLAB, Simulink
***Manufacturing & Machining***: Mill, Lathe, 3D Printer

## PROJECTS

***Bag Valve Mask Monitoring System*** *August 2025 - Present*

- Designed and fabricated a sensor-integrated attachment for a bag valve mask to monitor patient respiratory metrics.
- Integrated an embedded system to acquire and process real-time sensor data, interfacing pressure and temperature sensors with Bluetooth communication modules.
- Built a web-based dashboard to visualize real-time data with dynamic graphs, alerts, and system status indicators for clinical monitoring.

***PID-Controlled Ball Balancing Platform*** *November 2025*

- Designed a closed-loop PID control system to stabilize a steel ball to center of a two-axis tilting platform.
- Modeled system dynamics and tuned PID gains to minimize overshoot, settling time and steady-state error.
- Validated controller performance with MATLAB Simulink and physical testing.

***Stack & Sort Robot*** *April 2025*

- Prototyped and assembled an autonomous robot capable of detecting, grasping, and stacking colored blocks.
- Implemented motor control using PWM signals to drive stepper and servo motors via Raspberry Pi GPIO.
- Integrated computer vision using HSV color space detection for real-time object recognition and sorting.

***Hanging Crane with 2-Stage Pendulum*** *November 2024*

- Constructed a hanging crane prototype incorporating a two-stage pendulum to analyze system dynamics and control behavior.
- Developed a simulation model in MATLAB/Simulink and validated system performance by comparing simulated results with experimental prototype data.

## EXPERIENCE

**Sophomores Leading on Promoting Equality** – College Station, TX
*Service and Outreach Director* *March 2025 - Present*

- Collaborated with the Fundraising Committee to support philanthropy efforts benefiting *Health For All*, including organizing and managing an annual fundraising event.
- Partnered with local organizations and charities to expand outreach and increase community engagement.

**Texas A&M University** – College Station, TX *June 2025 - July 2025*
*Undergraduate Research, Department of Engineering Technology and Industrial Distribution*

- Redesigned and fabricated an open-source robotic hand, improving mechanical functionality and modularity.
- Reverse-engineered existing CAD models to develop modifiable design files in SolidWorks, enabling design customization and iterative prototyping.

**Robomaster's Engineer Team** – College Station, TX
*Team Member* *January 2023 - January 2024*

- Designed and integrated mechanical subsystems for competition robots, including gripper and conveyor mechanisms for object manipulation and transport.